

Similarity

Definition

For A and B , two $n \times n$ matrices, we say that A is similar to B if there is an invertible matrix P such that $P^{-1}AP = B$.

Fact: If A and B are similar then they have the same characteristic polynomial.

Suppose $P^{-1}AP = I$ then

$$\begin{aligned}\det(B) &= \det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) \\ &= \frac{1}{\det(P)} \det(n) \cancel{\det(P)} \\ &= \det(I)\end{aligned}$$

Note: $\det(P^{-1}) = \frac{1}{\det(P)}$

$$1 = \det(I) = \det(P^{-1}P) = \det(P^{-1}) \det(P)$$

If $B = P^{-1}AP$ then.

$$\begin{aligned}\text{char poly of } B & \text{ is } \det(B - \lambda I) = \det(P^{-1}AP - \lambda I) \\ &= \det(P^{-1}AP - \lambda P^{-1}P) \\ &= \det(P^{-1}(A - \lambda I)P) \\ &= \cancel{\det(P^{-1})} \det(A - \lambda I) \cancel{\det(P)} \\ & \quad \uparrow \\ & \quad \text{char. poly of } A.\end{aligned}$$

Conclusion: Any two similar matrices have the same eigenvalues.

Ex: $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ eigenvectors are 4 and 2

with eigenvectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Claim: A is similar to $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} = B$

$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 4 & -2 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 4 & -2 \end{pmatrix}$$

$$P^{-1}AP = B$$

One little problem ☹

Ex: $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ Char poly of A ?

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ -1 & -\lambda \end{pmatrix} = \underline{\underline{\lambda^2 + 1}}$$

A has no(?) eigenvalues?

The complex numbers

- Introduce a new quantity, i , such that $i^2 = -1$.
- The complex numbers are then all expressions of the form $a + bi$ where a and b are real numbers.

Operations on the complex numbers

- Addition:

$$(a + bi) + (c + di) = (a + c) + (b + d)i \quad |$$

- Multiplication:

$$(a + bi) \cdot (c + di) = (ac - bd) + (ad + bc)i \quad |$$

$$(bi)(di) = bd \underline{i^2} = -bd$$

$$(1 + \underline{i}) + (3 - \underline{i}) = 4 + 0i = 4$$

$$\begin{aligned} (1 + i)(3 - i) &= 3 + 3i - i + \underline{i(-i)} \\ &= 3 + 2i + 1 = 4 + 2i \end{aligned}$$

What about division?

If we have $\underbrace{(a+bi)}_{\downarrow}$, what is $\frac{1}{\underline{a+bi}}$?

$\neq 0 \ (0+0i)$

$$\left(\frac{1}{a+bi} \cdot \frac{a-bi}{a-bi} \right) = \frac{a-bi}{a^2+b^2} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i$$

the mult. inverse
of $a+bi$

$$(a+bi)(a-bi) = a^2 - abi + abi + b^2$$

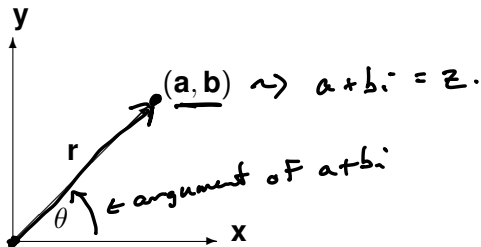
$$= a^2 + b^2$$

$$\text{Ex: } \frac{1}{i} = \frac{1}{i} \frac{-i}{-i} = \frac{-i}{1} = -i$$

$$(\text{Notice: } i(-i) = 1)$$

The complex plane

$$r = \sqrt{a^2 + b^2}$$
$$= \sqrt{z \cdot \bar{z}}$$

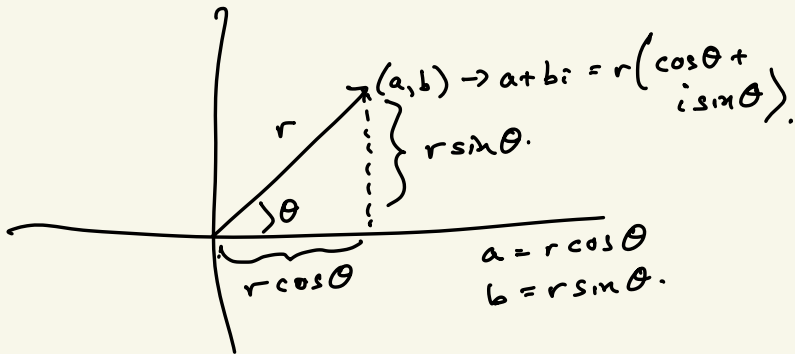


- $r = \sqrt{a^2 + b^2}$; this is called the modulus of the complex number $z = a + bi$ and written $|z|$.
- We saw that $z \cdot \bar{z} = |z|^2$.
- θ is an argument for $a + bi$ and is only determined up to multiples of 2π .
- $a = r \cos(\theta)$ and $b = r \sin(\theta)$ so $z = r(\cos(\theta) + i \sin(\theta)).$

$$z = a + bi$$

$\bar{z} = a - bi$, the conjugate of z .

$$z\bar{z} = a^2 + b^2$$



A beautiful formula and a caveat going forward

- Exponentiation can be defined on complex numbers as follows:

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

- Consider then that if $\theta = \pi$ then

$$e^{i\pi} + 1 = 0$$

Linear algebra over the complex numbers

Everything we have done in the course to date - linear systems, matrices, determinants, eigenvalues, diagonalization - all goes through **unchanged** if we are using complex numbers instead of real numbers. Going forward it will be considered fair to have complex entries in matrices and to consider complex roots of polynomials.

Multiplication and division in polar form

- If

$$z_1 = r_1(\cos(\theta_1) + i \sin(\theta_1)) \text{ and } z_2 = r_2(\cos(\theta_2) + i \sin(\theta_2))$$

- then $z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$ and
- if $z_2 \neq 0$ then $\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2))$.
- A special case occurs when you compute z^n where $z = r(\cos(\theta) + i \sin(\theta))$:

$$z^n = r^n (\cos(n\theta) + i \sin(n\theta))$$

- We can also compute n^{th} roots of $r(\cos(\theta) + i \sin(\theta))$; they are

$$r^{1/n} \left(\cos\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) + i \sin\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) \right) \text{ for } k = 0, 1, \dots, n-1$$